What Makes Small Decision Models Transfer? Schema Augmentation and Data-Mixture Trade-Offs in a 350M Decision Model

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***Abstract*—**

***Keywords*—System One AI, decision models, Liquid Foundation Models, probability calibration, edge inference**

I. Introduction

Generative autoregressive language models are increasingly deployed for structured enterprise decisions: payment fraud routing, customer inquiry triaging, and policy enforcement. In this setup the model is prompted to emit its decision as a formatted payload such as JSON or XML. This *System Two* paradigm carries three operational costs. First, latency: emitting a formatted decision requires $O(T)$ sequential decoding steps, spending GPU cycles on boilerplate punctuation while holding the key-value cache. Second, syntactic reliability: even with constrained decoding, autoregressive sampling retains a non-zero probability of malformed output. Third, epistemic pathology: preference-based post-training rewards confident answers, so raw output scores are systematically overconfident and unusable for risk-controlled automation [1].

System One decision models separate structured judgment from text generation: an input state S and a typed question schema Q are mapped directly to a joint probability distribution over candidate options in a single forward pass ($O(1)$ decoding steps), with no free-form text and no output parsing. The paradigm was pioneered by TypeSafe Jev (closed API), Cloudflare Clef [2], Convai Laya, and the open Unsloth training recipe [3]. Existing work concentrates on dense bidirectional encoders or decoder-only transformers; hybrid convolution-attention backbones at sub-billion scale, attractive for edge deployment, remain an unexplored substrate for decision heads. We study LiquidAI LFM2.5-350M [4], a 350M hybrid model combining gated-convolution blocks with grouped-query attention.

Rather than proposing a new architecture (our decision head follows the Clef design), this paper asks which *training recipe* choices make small decision models transfer: (RQ1) does schema augmentation (option renaming, description paraphrasing, derived questions) produce transfer to unseen decision taxonomies; (RQ2) how does data-mixture composition trade off across task groups under fixed 350M capacity; with calibrated probabilities and selective-classification coverage evaluated throughout.

Our contributions are: (1) an adaptation recipe and open artifact, `itsdevice/lfm-2.5-350m-decs1`; (2) controlled experiments — multi-seed runs, leave-one-taxonomy-out evaluation, and token-matched augmentation ablations [PENDING run #1–8, tracker docs/RECOMMENDATION.md §6.4]; (3) a calibration evaluation (proper scoring rules plus post-hoc temperature scaling) with selective coverage at fixed risk; and (4) a capacity-frontier analysis against a 0.8B dense baseline run under the identical protocol [PENDING run #8]. This is an empirical systems paper: every comparative claim traces to a logged run, and result numbers enter this section only after the tracker turns green.

The remainder of the paper covers related work, the architecture and training methodology, the dataset composition and augmentation pipeline, the experimental results, and deployment discussion with limitations.

II. Related Works

III. Dataset and Preprocessing

A. Dataset Description

B. Feature Selection

C. Data Preprocessing

IV. Proposed Methodology

A. Overall Framework

B. Machine Learning Models

1) Random Forest Regression:

2) Extreme Gradient Boosting (XGBoost):

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C. Genetic Algorithm for Hyperparameter Optimization

D. Explainable AI using SHAP

V. Experimental Setup

A. Evaluation Metrics

B. Implementation Details

VI. Results and Discussion

A. Performance Comparison

B. Computational Cost Analysis

C. Explainability Analysis

VII. Conclusion

References

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